

Morteza Shoushtari

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PROFESSIONAL SUMMARY

As a Senior Network/System Engineer at Supermicro, I specialize in AI and HPC network infrastructure, with deep expertise in wired, wireless, and cellular networking technologies. I hold a Ph.D. in Computer and Electrical Engineering, with a research focus on the network and secrecy/error-correction coding. Passionate about working with great teams and always learning in fast-moving technical environments.

SKILLS

Wired/Wireless/Data Center Network Architecture, Infrastructure, and Protocols | AI/HPC network design | InfiniBand/Ethernet Networking | Cellular Networks Technologies (LTE/5G) | Protocol Stack (PHY, MAC, RLC, PDCP, RRC, SDAP) | TCP/ IP | InfiniBand, Ethernet | RDMA, RoCE | Fiber optics/transceivers | 802.11 (WiFi), 802.15 (Bluetooth, Zigbee) | IP Networking | IPv4 and IPv6 | FLSM and VLSM | L2 Switching (VLAN, STP, LACP) | L3 Routing (EIGRP, OSPF, BGP, MPLS) | Server Operating System (Mac, Windows, Linux) | Virtualization Technologies | Network Troubleshooting Analysis (Wireshark) | Process Automation | Cryptographic Algorithms (AES, DES, RSA) | Post-Quantum Cryptosystems | Error-Correction Coding (LDPC, Polar, Turbo, Viterbi) | Physical Layer Security

Programming Languages: C++, MATLAB

Certificates: Supermicro GPU Service Associate/Specialist/Expert (SGSA, SGSS, SGSE), Supermicro Direct Liquid Cooling Specialist, Nvidia Infiniband, CCNA, HCNA, MCITP, Network+, A+

WORK EXPERIENCE

Super Micro Computer, Inc.

CA, USA

Sr. Network/System Engineer

May 2024 – Present

- Design, optimize, and validate network infrastructure for large-scale AI/HPC systems, including NVIDIA (GB300 and VR), and AMD (Helios) architectures across InfiniBand and Ethernet fabrics.
- Develop and optimize port connectivity by shuffle solutions (cables/modules), cabling architectures, and high-speed interconnects to ensure optimal performance, scalability, and signal integrity.
- Lead network and system validation activities, developing test plans, validation procedures, and technical documentation for network rack components.
- Collaborate with cable, transceiver, and switch vendors to troubleshoot and resolve complex system and network issues.

Brigham Young University

Utah, USA

Graduate Research Assistant

Jan 2019 – July 2023

- Discovered new properties of secrecy coding from the perspectives of information and coding theory.
- Devised a high-speed algorithm to construct the optimal code in Nested Linear Secrecy Codes, enhancing code identification and selection efficiency.
- Conducted a detailed assessment and visualized secrecy problems such as eavesdropping in aeronautical mobile telemetry communication and suggested the use of secrecy coding and specialized version of post-quantum cryptosystems for this type of communication systems.

HUAWEI

Tehran, Iran

Network Engineer

Aug 2016 – Aug 2017

- Designed and implemented wired/wireless networks, and performed network maintenance and system upgrades.
- Maximized network performance and increased network availability by 99.99% by implementing a redundant network architecture.

- Performed installations and configurations, and support of various network devices such as switches (Huawei, Cisco), wireless access points (Huawei, Aruba), cameras, and video conferencing systems.

JYAN

Network Engineer

Tehran, Iran

Jan 2016 – Aug 2016

- Managed wired communication networks, and performed network maintenance and system upgrades.
- Monitor performance and ensure system availability and reliability.
- Performed installations and configurations of various network devices such as switches (Cisco), wireless access points (Aruba), cameras, and video conferencing systems.

Persia Cloud

Network Engineer

Tehran, Iran

Jun 2015 – Jan 2016

- Performed installations and configurations of HP servers, virtual machines (VMs), and Cisco switches.
- Administered SaaS and PaaS layers of the company's cloud platforms (Citrix, IIRAS).
- Managed and configured Microsoft application servers CRM, SharePoint, and Lync.

EDUCATION

Brigham Young University

Ph.D. in Electrical and Computer Engineering; GPA: 3.78/4.00

UT, USA

Jan 2019 – July 2023

Shiraz University

M.Sc. in Information Technology; GPA: 3.74/4.00

Fars, Iran

2011 – 2013

SELECTED PUBLICATION

- “Secrecy Coding for the Binary Symmetric Wiretap Channel via Linear Programming”, IEEE Transactions on Information Forensics and Security, 2025.
- “Optimizing Finite Blocklength Nested Linear Codes: Using the Worst Code to Find the Best Code”, Entropy Journal, 2023.
- “Towards Practical Physical-Layer Security: Channel Measurements and Pedestrian Traffic”, under review IEEE Transactions on Information Theory, 2023.
- “Classification of Coset Codes for Wiretap Channels”, under review IEEE Transactions on Information Theory, 2023.
- “From Privacy Protection to Analyzing Users’ Behavior: The Crucial Role of Information Theory in the Metaverse”, IEEE Inter-mountain Engineering, Technology, and Computing Conference, 2023.
- “A Comparative Study of Waveforms Across Mobile Cellular Generations: From 0G to 5G and Beyond”, under review IEEE Access, 2023.
- “Post-Quantum Cryptography Based on Codes: A Game Changer for Secrecy in Aeronautical Mobile Telemetry”, in Proc. of the International Telemetry Conference (ITC), Las Vegas, NV, US, Oct. 2022.
- “Secrecy coding in the Integrated Network Enhanced Telemetry (iNET)”, in Proc. of the International Telemetry Conference (ITC), Las Vegas, NV, US, Oct. 2021.
- “New Dual Relationships for Error-Correcting Wiretap Codes”, in Proc. of IEEE Inform-ation Theory Workshop (ITW), Kanazawa, Japan, Oct. 2021.
- “On Caching with Finite Blocklength Coding for Secrecy over the Binary Erasure Wiretap Channel”, in Proc. of IEEE Wireless Telecommunications Symposium (WTS), San Francisco, US, Apr. 2021.

AWARDS & ACHIEVEMENTS

- Best paper award, International Telemetry Conference, 2022.
- Second best graduate student paper award, International Telemetry Conference, 2021.
- Third place award, IEEE Intermountain Engineering, Technology, and Computing Conference (IEEE i-ETC), 2023.
- Outstanding IT engineer in Huawei Technologies Company, 2017.

REFERENCES

Available upon request.